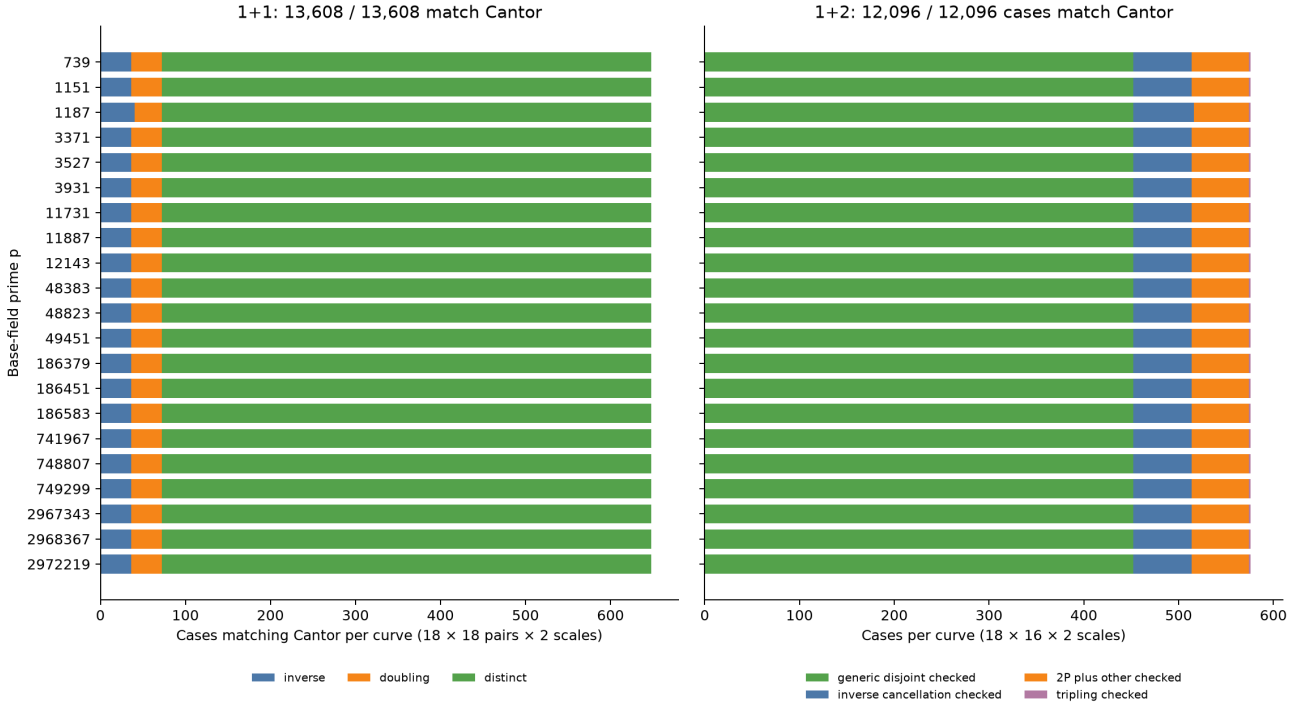
Published genus-2 projective law: sampled branch checks

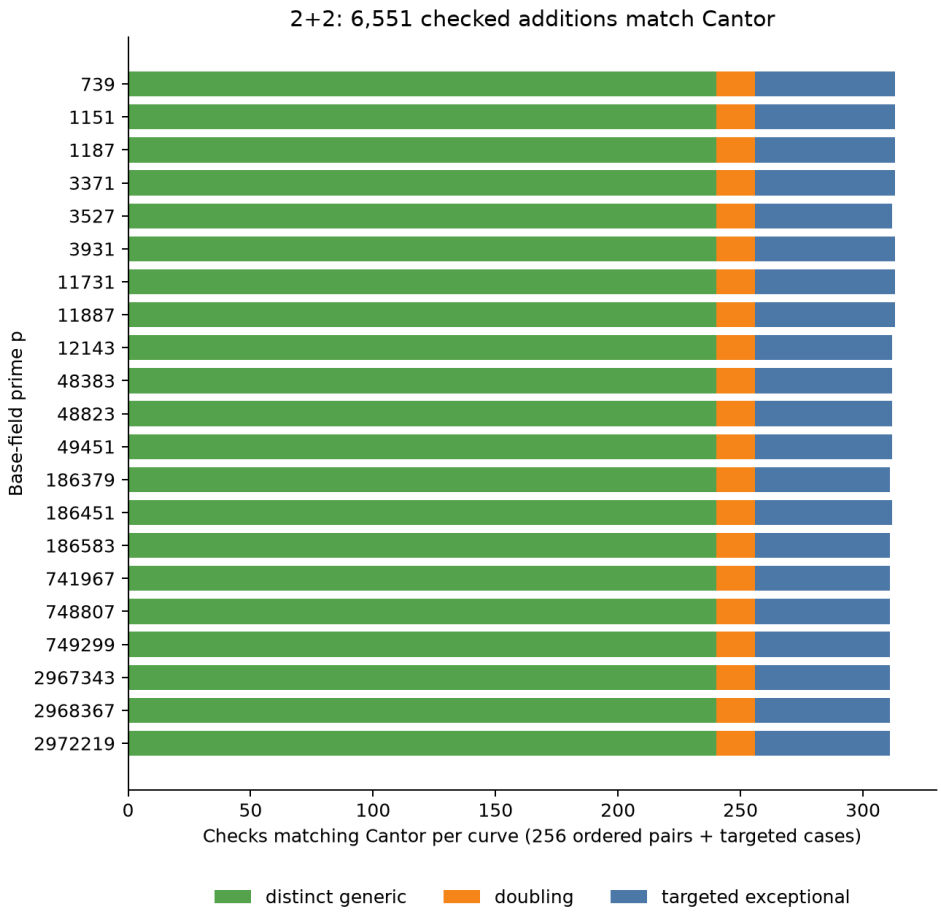
Reproducible experiment report

21 October 2026

The independently implemented degree-one plus degree-one formulas are equations (13) and (14); the degree-one plus degree-two branches use equations (15), (16), (17) and (18) of Ozbay Gurler–Hisil (2024).

All 21 normalized Scholten models were checked at two projective scales; each scale used 18 rational points in ordered pairs. All 13,608 outputs matched an independent Cantor implementation after boundary normalization: 760 inverse pairs, 752 doublings and 12,096 distinct-point additions. All 12,096 sampled 1+2 cases matched Cantor: 9,492 generic disjoint, 1,304 inverse-support cancellations, 1,258 repeated-support additions with a distinct remaining point, and 42 triplings. The 2+2 sample comprises 5,040 generic additions, 336 ordinary doublings, 483 degree-drop additions, 336 inverses, 294 shared-support additions, 42 same-polynomial mixed-sign additions, 13 special-point doublings and 7 degree-drop doublings. All 6,551 evaluated outputs matched Cantor. The targeted exceptional cases are unevenly distributed across models; this finite sample does not establish exhaustive branch coverage. The native executable was rerun and the 21 JSONL records matched byte-for-byte. This is a branch-level correctness result. Any unexercised 2+2 branches, the full DLP pipeline and its 1,260 paired rho controls have not run under this law.





Raw records: `results/complete-law-branches.jsonl`. Source: `src/bin/complete_law_gate.rs`. Arithmetic reference: Cantor addition in the repository's independently developed Jacobian module.